



Parasitic dinoflagellate *Hematodinium* in marine decapod crustaceans: a review on current knowledge and future perspectives

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Abstract

Parasitic dinoflagellates of the genus *Hematodinium* are known to infect various marine crustaceans worldwide, especially crabs and several species of shrimp and lobster. Some of these species are new host species and components of commercial fishery products. These parasitic species are predominantly found in the hemolymph of the host and cause pathological changes and functional damage to organs and tissues, leading to death. In recent years, these parasites have infected important commercially valuable species, particularly in European waters, US waters, Australian waters, and recently in Shandong Peninsula in China. These *Hematodinium* pathogens were also reported to affect wild shrimp in Chinese waters and in the English North Sea. These rapid spreads affect crustacean aquaculture industries, where they are indeed a significant threat to the sustainability of the aquaculture of important crustaceans. The fishery products industries are also under pressure from the invasion of this pathogen, as the crab meat produced has a bitter taste, which may reduce its marketability. In response to these threats, this review was aimed at providing a broader understanding of the development of parasite distribution and ecological aspects of *Hematodinium*. In addition, the interaction of these pathogens with their hosts, the environmental drivers of *Hematodinium* disease, and future research perspectives were discussed.

Keywords Crustacean · Dinoflagellates · Disease · Economic losses · *Hematodinium*

Introduction

Crustaceans are among the most widely consumed seafood species globally due to their high nutritional content, distinctive flavors, and use in a variety of cuisines (Farag et al. 2023). They are a large group of arthropods with the fourth greatest diversity among other animal groups and are usually

considered a subphylum of the phylum Arthropoda (Susanto 2021). They are a large group of arthropods with the fourth greatest diversity among other animal groups and are typically regarded as a subphylum of the arthropods (Susanto 2021). Until now, there have been 67,000 different species of crustaceans in 6 classes, and many of them are free-living aquatic animals (Ovchinnikova and Shi 2023). Some of them are well-known marine animals such as lobsters, crabs, shrimp, krill, and barnacles, and they are commonly threatened by the presence of various forms of parasites that frequently use them as hosts and are often invasive (Laguer 2017). Parasites are components of the aquatic ecosystem that generally have negative effects on the host's health through several detrimental effects and stressors that may cause mortality. The presence of parasites in hemolymph crustaceans will slowly disable the host's respiratory functions, disrupt the activity of biochemical enzymes, and consume nutrients from the host, eventually leading to host death (Li et al. 2021).

Several studies have reported the existence of parasites that invade the hemolymph of crustaceans including

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Hematodinium (Chatton and Poisson 1931); (Gallien 1938); (Hudson and Shields 1994), *Pekinsus* sp. and *Blastodinium* sp. (Nzeako et al. 2015)), *Paramikrocytos canceri* (Edwards et al. 2019), *Haplosporidium* sp. (Davies et al. 2020), *Lynnia grapsolytica* (Metz and Hechinger 2021), *Toxoplasma gondii* and *Cyclospora cayetanensis* (Marangi et al. 2022), *Mesanoophrys* sp. (Lei et al. 2022), and microsporidia (Bojko et al. 2018). Their presence in the hemolymph results in severe pathologies for the host, like organ failure and respiratory dysfunction (Li et al. 2008), which can lead to death. In this current review paper, our work aimed to provide a broader understanding of a group of hemolymph parasites belonging to the genus *Hematodinium*, which includes their current taxonomic evaluations, lifecycle, distributions, and nutrition. Moreover, the interaction between this *Hematodinium* parasite and its host, the gross signs, and the pathogenic repercussions of parasitic diseases were also covered. Besides, the environmental drivers of disease and the potential economic damages to crustacean production were also discussed. Finally, the mode of disease transmission and the potential control methods will also be highlighted.

Hematodinium

Taxonomy

Dinoflagellates are a unique group of organisms because they are closely related to Ciliophora and Sporozoa (Apicomplexa), which places them in a superphylum (Gajadhar et al. 1991); (Schnepf and Elbrächter 1992). Out of more than 2400 known dinoflagellate species that have been documented, 140 species in 35 genera are known to be parasites (Rowley and Coates 2023). Pouchet (1884) were the first to describe the dinoflagellate parasite in the intestine of tunicates, named *Gymnodinium pulvisculus*. Several studies have found that dinoflagellate parasites infect a wide variety of hosts, including fish (Brown 1934), algae (Gachon et al. 2010), annelids (Yamamoto et al. 2020), rotifers (Gómez and Skovgaard 2015), and crustaceans (Purushothaman et al. 2020). In 1964, Chatton's student classified dinoflagellates into two subdivisions: the ecto-parasite Blastodinia (subdivision Dinokaryota) and the internal parasitic Duboscquodina (latest Syndinea) (Cachon 1964). Blastodinales such as *Blastodinium* sp. invade the intestinal tract of marine copepod hosts, whereas Duboscquodina (Syndinian) like *Hematodinium* impact several invertebrates, particularly decapods (Gong et al. 2023); (Shields 1994)). Now, Syndinea subdivision holds the position occupied by *Hematodinium*. The inhabitants of this subdivision are all classified as parasites and only have members of one class, namely Syndiniophyceae of the order Syndiniales. *Hematodinium*

are classified under the genus *Syndinium* in the family Syndiniaceae (Chatton and Poisson 1931); (Fensome 1993).

Hematodinium can be identified as dinoflagellates by the characteristic V-shaped nuclei, alveolate pellicle, presence of naked gymnodinoid dinospores (or zoospores), and the presence of classic mitosis (dynamitosis) with regular mitotic activity in the nucleus (Messick 1994; Stentiford and Shields 2005). There are 25 species of parasitic dinoflagellates that have been identified and are found in two orders: Blastodinales (subdivision Dinokaryota) and Syndinales (Shields 1994). Four genera of Syndinales are known as parasites on crustaceans, including *Actinodinium* (copepoda), *Trypanodinium* (copepod eggs), *Syndinium* (crustacean eggs), and *Hematodinium* (decapoda crustacea) (Shields 1994). However, there has not been enough research done on *Actinodinium* (in copepods) and *Trypanodinium* (in copepod eggs) (Shields 1994), possibly because *Actinodinium* parasites are still very limited since they were documented from the stomach wall of the copepod *Acartia clausi* (Chatton and Hovasse 1938); (Rose and Cachon 1951); however, *Trypanodinium* has never been documented again after being described from the eggs of several copepod species at the beginning of the twentieth century.

Due to several obstacles, including the need for comparative studies on understanding the lifecycle and distinguishing parasite features, to date only two species of *Hematodinium* have been described: *Hematodinium perezii* and *Hematodinium australis*. These two species are distinguished by their trophal sizes (vegetative stage), their austral locations, the presence of a globular plasmodial stage, and a tiny beaded form of condensed chromatin. In the host, their presence in hemolymph is identical to that of the high-intensity amoeboid trophont, dinokaryon, and filamentous plasmodial phases. After identification using morphology, more intricate and thorough introductions were initiated in the 1990s through molecular study. For *Hematodinium*, the first application of PCR technology was conducted by (Hudson and Adlard 1994), where they used *Cancer sapidus*, *C. opilio*, *C. bairdi*, *Nephrops norvegicus*, and *Petrolisthes armatus* infected with *Hematodinium*. The primers were bound at the 3' end of the gene from the small subunit rDNA region (SSU rRNA) and the 5.8S rRNA gene to amplify a 680-bp fragment of *Hematodinium* that infected animal samples. Next, (Hudson and Adlard 1996) performed an update by sequencing the 3' end of the SSU rRNA gene as well as the 5' end of the rRNA from test animals. Hence, there are substantial sequence differences between different *Hematodinium* species, including *C. sapidus*, *N. norvegicus*, and *C. bairdi*.

Meanwhile, from a regional phylogenetic point of view, (Small et al. 2012) discovered three genotypes of *H. perezii*: genotype I was found in *Liocarcinus depurator* in the English Channel, genotype II in *Portunus trituberculatus* and *Scylla paramamosain* in China, and genotype III in *C.*

sapidus in North America. Hence, the author claimed that the PCR primer set and region phylogenetics are two of the DNA-based molecular diagnostic procedures that can be utilized to diagnose and offer verification of *Hematodinium* infection. Until now morphological studies, molecular evidence, diagnostic techniques, exploration of other host species, and biogeography of *Hematodinium* have helped researchers and fisheries practitioners to understand better this *Hematodinium* parasite and the diseases it causes.

Life cycle

Although the complete *Hematodinium* lifecycle is yet unknown, it has been noted in a number of studies that involve either in vivo or in vitro processes. According to (Eaton et al. 1991), the in vivo life stage comprises at least three phases, including trophont or plasmodia (i.e., vegetative or growth stage), sporont (i.e., non-vegetative stage), and the infectious agent dinospore (i.e., mastigote or motile stage). During the vegetative stage, mononuclear trophonts and plasmodia can be seen under a light microscope to have multinucleated nuclei containing “foamy” and basophilic cytoplasm (Li et al. 2011); (Ryazanova et al. 2016), and trophonts often develop in the hepatopancreas and also in other organs (Ni Chualáin and Robinson 2011). Then, the sporonts emerge from the trophont and are formed as sporoblasts, which then develop into prespores, which are transitional stages before becoming dinospores (Li et al. 2011); (Wang et al. 2017). The last stage of *Hematodinium* in vivo is the dinospore, which is present with a motile diffusing form of the parasite (Li et al. 2011). There are two types of dinospores (i.e., macrodinospores and microdinospores), which are classified based on size and morphology, and these dinospores can exit the host through the mouth and gills (Shields and Squyars 2000) or through the gills alone and cause sporulation, which then circulates in the aquatic environment (Huchin-Mian et al. 2018). In an aquarium environment, sporulation is usually marked by a change in the color of the water in the host’s aquarium (Shields and Squyars 2000). Apart from morphology and shape, these two dinospores also differ in terms of their functional behaviors, where microspores swim fast to accelerate the spread of infection (Chen et al. 2023).

Meanwhile, the in vitro life cycle of *Hematodinium* was found in various species of crustaceans (Li et al. 2011); (Gaudet et al. 2015); (Wang et al. 2017). Generally, the life cycle was observed progressing through various developmental stages in a blue crab host. (Li et al. 2011) gave an understanding that the initial stage of infection in a new host begins in the filamentous trophont, which is commonly found in hemolymph hosts with mild infection. Thereafter, they develop into the amoeboid trophont and then the

arachnoid trophont. Arachnoid trophonts were the main proliferative phase of the *Hematodinium* in vitro, which usually occurs in the hepatopancreas and heart. Then the arachnoid trophonts develop as a mass within the arachnoid trophont, which gives rise to sporoblasts (i.e., a single cell or a cellular mass that is usually found in the hemolymph of chronically and heavily infected hosts). Prespores are the intermediate phase of sporoblasts and dinospores and are rarely seen in chronic and heavily infected hemolymphs. Motile dinospores are the late phase of the life cycle and consist of micro- and macrospores, which are commonly found in the water column and sometimes in the host’s hemolymph. In addition, a pansporoblast phase developed by arachnoid sporonts seems to be another alternative for parasite proliferation in the host. This phase then undergoes segmentation to form gorgonlock colonies, which in turn become clump colonies in the form of sporoblast aggregates, and they will then develop into arachnoid trophonts during in vitro subculture. In vitro *Hematodinium* from *P. trituberculatus* was also reported by (Wang et al. 2017) in their laboratory, which generally showed results that were relatively the same as previous studies (Li et al. 2011); (Gaudet et al. 2015). Based on the description above, it is known that *Hematodinium* in vivo shows few life stages compared to *H. perezii* in vivo and in vitro. The life cycle phases obtained in vitro do not appear to be representative of the natural life phases of *Hematodinium*. This could be because in vitro natural tissue structures are absent, so some phases of life in in vitro culture may result from artificial conditions. In addition, when *H. perezii* infects *C. sapidus*, *Hematodinium* sometimes adheres to the walls of the hemolymph vessels in the form of multiple nuclei (Small et al. 2012). Therefore, complex stages in vitro may occur in vivo, especially during clinically asymptomatic infections. The life stages observed in *Hematodinium* isolates can also differ between host species and environmental conditions because of the parasite’s ability to adapt to the host or change according to environmental conditions.

Distribution and immunity

Crustacean hemolymph has been documented to be infected by a number of parasites. *Hematodinium* was first identified by (Chatton and Poisson 1931) from *Carcinus maenas* and *L. depurator* collected from the French coastline, with a low prevalence (4 of 3570 infected crabs). (Gallien 1938) reported the spread of *H. perezii* in *Portunus latipes*. In the south eastern USA, for the first time, *H. perezii* was documented in a very valuable commercial crab (i.e., blue crab, *Callinectes sapidus*). Then, in the Mid-Atlantic, it was found in two cancer crab species, *Cancer irroratus* and *Cancer borealis*, as well as other portunids, including *Ovalipes ocellatus* (MacLean and Ruddell 1978). Meanwhile

H. australis was first described to infect the Australian sand crab, *Portunus pelagicus* (Hudson and Shields 1994). In Asia, the invasion of *Hematodinium* was first documented in *Scylla serrata* from the coastal eastern region of China (Xu et al. 2007a). For the lobster and shrimp groups, *Hematodinium* was first reported by (Field et al. 1992), who reported that it infected *N. norvegicus* sp. in Scotland. For regions outside Europe, it was first documented by (Xu et al. 2010) on the shrimp *Exopalaemon carinicauda* in the Islands of Zhoushan, China. So far, there have been 53 species of crustaceans, and detailed illustrations are tabulated in Table 1 to shown current findings of prawn, lobster, and crab, which have been infected with *Hematodinium*.

According to (Small 2012), there will be more host species and new types of parasites as long as there are still economically viable crustacean species that are caught in the wild and raised on farms. Moreover, new pathogens and hosts are always discovered as a result of the distribution of parasites through ocean currents, host movements, range extension, handling mechanisms, transportation (e.g., shipping, cultivation), and research activities (Ryazanova et al. 2021). The distributions of *Hematodinium* around the world are tabulated in Fig. 1.

In terms of immunity, the relationship between the immune system and the existence of *Hematodinium* in the host is still poorly understood. Previous reports revealed that the invasion of the gazami crab (*P. trituberculatus*) by *H. perezii* on a laboratory scale showed that the host reacts to the parasite at the mRNA and protein level, with fluctuations in the protein expression of immune candidates including pro-phenoloxidase (Li et al. 2015; Li et al. 2019); 2022); however, this finding contradicted (Coates et al. 2023) and (Rowley et al. 2015), who found that *Hematodinium* invasion caused no immune suppression (e.g., phagocytosis, encapsulation, or degranulation) in the tested crabs. Although (Li et al. 2019, 2022) have presented research on the OMICS technique in infected crabs and stated that there were changes in the main pathways and proteins involved in host-parasite interactions in their study, they did not use negative controls (i.e., crabs with no parasite) rather changes that occurred after — challenges with *Hematodinium*, so the basis for comparison in the research is still lacking, creating a gap for further research to use control to increase validity and reliability. Thus, more accurate conclusions regarding host–pathogen interactions can be made. However, these findings conflict with (Coates et al. 2023) and (Rowley et al. 2015), who found that *Hematodinium* invasion did not cause immune suppression (e.g., phagocytosis, encapsulation, or degranulation) in the crabs tested. According to (Coates et al. 2023), *Hematodinium* can impair hemocyte communication and reduce extracellular vesicles in the host when compared to control crabs. Cellular immune defenses are

assumed to stop functioning as a result of this. Although *Hematodinium* is a parasite that mostly inhabits the hemolymph, the report nonetheless poses an important scientific query regarding how *Hematodinium* can alter the immune response, consequently impeding the optimal function of hemocytes. This problem presents a fresh viewpoint that merits additional scholarly investigation.

Nutrition

Pure *Hematodinium* is heterotrophic and lacks chloroplasts, making it incapable of self-sustenance. It lives and develops itself within the hemolymph or body cavity of a host and obtains the energy and nutrients it needs through osmoth-erapy (Shields 1994), which causes hosts to decrease their glycogen concentrations in the muscles and also a drop in glucose levels (Stentiford et al. 2001a). (Shields et al. 2003) reported that the infected female crabs dropped their glycogen levels in the hepatopancreas to 50%, while for males they dropped to 70%. The hepatopancreas of crustaceans plays an important role in nutrition by facilitating the absorption of digested food products, the production of mineral and organic reserves (Eizen and Jeffrey 1969), the metabolism of lipids and carbohydrates (Ceccaldi 1989), as well as storing nutrients for use by the body's tissues (Vogt 2019). Besides, the function of glycogen in muscles is to transport glucose from the hemolymph to the muscles and hepatopancreas (Duan et al. 2022). Glycogen depletion in the muscles and hepatopancreas is thought to be a manifestation of the host response to stressful environmental or physiological conditions. The depletion of glycogen in the hepatopancreas due to parasite infection may also lead to growth retardation. Lipids and glycogen are also used during molting, fasting, and reproduction (Ciaramella et al. 2014). Glycogen depletion in the muscles and hepatopancreas is thought to be a manifestation of the host's response to stressful environmental or physiological conditions. Taking into account the crucial role of glycogen and hepatopancreas in host growth, as well as additional information from (Ciaramella et al. 2014), who stated that lipids and glycogen are also utilized during the molting, fasting, and reproduction, it can be assumed that glycogen depletion can potentially inhibit host growth. Although there are no specific studies examining the relationship between glycogen depletion as a result of *Hematodinium* parasite invasion and host growth, (Wu et al. 2022) showed that the decrease in glycogen caused by *Enterocytozoon hepatopenaei* consumption as well as deregulation of lipid metabolism played a role in inhibiting the growth of *Litopenaeus vannamei* shrimp. Therefore, it is highly recommended to develop studies on hosts infected with *Hematodinium*. Through this, preventive

Table 1 List of decapod crustaceans that hosted *Hematodinium* spp. in the hemolymph

Host	<i>Hematodinium</i> species	Season/month; reported places	Morphological methods used	Reference (s)
Prawn				
<i>Exopalaemon carinicauda</i> sp.	<i>Hematodinium</i> sp.	NM; Zhejiang, China	HS,H,U,I,PCR,S	Xu et al. (2010)
	<i>Hematodinium</i> sp.	NM; Zhejiang, China	PCR,S	Xiao et al. (2016)
<i>Crangon crangon</i> sp.	<i>Hematodinium</i> sp.	NM; North Sea, UK	MM,H,U,PCR,S	Stentiford et al. (2012)
Lobster				
<i>Nephrops norvegicus</i> sp.	<i>Hematodinium</i> sp.	Summer; West Coast of Scotland	MM,HS,H,U	Field et al. (1992)
	<i>Hematodinium</i> sp.	NM; Clyde Sea Area, Scotland	MM,H,U	Field and Appleton (1995)
	<i>Hematodinium</i> sp.	April and June; Firth of Clyde, Scotland	MM	Taylor et al. (1996)
	<i>Hematodinium</i> sp.	NM; Clyde Sea Area, Scotland	HS,U	Appleton and Vickerman (1998)
	<i>Hematodinium</i> sp.	Spring; Skagerrak, Sweden	MM	Tärnlund (2000)
	<i>Hematodinium</i> sp.	Spring; Clyde Sea Area, Scotland	MM	Stentiford et al. (2001a, 2001b)
	<i>Hematodinium</i> sp.	NM; Clyde Sea Area, Scotland	MM, I	Small et al. (2002)
	<i>Hematodinium</i> sp.	Spring; Western Irish Sea, Ireland	MM	Briggs and McAliskey (2002)
	<i>Hematodinium</i> sp.	NM; Clyde Sea, Scotland	I, PCR	Small et al. (2006)
	<i>Hematodinium</i> sp.	NM; Irish Sea, Ireland	PCR	Small et al. (2006)
	<i>Hematodinium</i> sp.	NM; Skagerrak, Sweden	PCR	Small et al. (2006)
	<i>Hematodinium</i> sp.	NM; Kettegat, Denmark	MM,PCR,S	Eigemann et al. (2010)
	<i>Hematodinium</i> sp.	February; Clyde Sea Area, Scotland	I, PCR	Beevers et al. (2012)
Crab				
<i>Carcinus maenas</i> sp.	<i>H. perezii</i> sp.	NM; Roscoff, France	HP	Chatton and Poisson (1931)
	<i>H. perezii</i> sp.	NM; Arachon, France	HP	Chatton and Poisson (1931)
	<i>H. perezii</i> sp.	NM; Newton's Cove, Weymouth, UK	H,U	Stentiford et al. (2004)
	<i>H. perezii</i> sp.	NM; Southampton, UK	H	Stentiford and Shields (2005)
	<i>Hematodinium</i> sp.	NM; West Coast, Scotland	MM,H,PCR,S	Hamilton et al. (2007)
	<i>Hematodinium</i> sp.	NM; UK and Faroe Islands	MM,H,PCR,S	Bojko et al. (2018)
	<i>Hematodinium</i> sp.	Spring; Swansea Bay, UK	PCR,S,HP,H	Davies et al. (2019)
	<i>Hematodinium</i> sp.	NM; South Wales coast, UK	H,PCR,S,PG	Davies et al. (2022)
	<i>Hematodinium</i> sp.	NM; Swansea Bay, UK	HP, PCR	Coates et al. (2023)
<i>Liocarcinus depurator</i> sp.	<i>Hematodinium perezii</i> sp.	NM; Luc-ser-mer, France	HP	Chatton and Poisson (1931)
	<i>Hematodinium</i> sp.	Spring; Clyde Sea, Scotland	PCR,S	Hamilton et al. (2009)
	<i>Hematodinium</i> sp.	NM; Kettegat, Denmark	MM,PCR,S	Eigemann et al. (2010)
	<i>Hematodinium</i> sp.	NM; South Coast of England	H,U,PCR,S	Small et al. (2012)
<i>Portumnus latipes</i> sp.	<i>Hematodinium</i> sp.	NM; Wimereux, France	HP	Gallien (1938)
	<i>Hematodinium</i> sp.	NM; France	HP	Chatton (1952)
<i>Ovalipes ocellatus</i> sp.	<i>Hematodinium</i> sp.	NM; Roscoff, France	HP	Chatton (1952)
	<i>H. perezii</i> sp.	NM; New York Bight, USA	HP	McLean and Ruddell (1978)

Table 1 (continued)

Host	<i>Hematodinium</i> species	Season/month; reported places	Morphological methods used	Reference (s)
<i>C. sapidus</i> sp.	<i>Hematodinium</i> sp.	Late autumn; North Carolina, Georgia, and Florida, USA	HS,U	Newman and Johnson (1975)
	<i>H. perezi</i> sp.	Late autumn; coastal of Maryland and Virginia, USA	HS,H	Messick (1994)
	<i>H. perezi</i> sp.	Summer and autumn; Virginia, USA	HP,H	Shields and Squyars (1999)
	<i>Hematodinium</i> sp.	Late autumn; Atlantic and Gulf Coasts, USA	HP	Messick and Shields (2000)
	<i>Hematodinium</i> sp.	NM; Georgia, USA	PCR,S	Gruebl et al. (2002)
	<i>Hematodinium</i> sp.	Late spring and fall; Georgia, USA	HP,H,PCR,S	Sheppard et al. (2003)
	<i>H. perezi</i> sp.	October; Georgia, USA	PCR,S	Frischer et al. (2006)
	<i>Hematodinium</i> sp.	NM; Delmarva Peninsula & Chesapeake Bay, USA	H,PCR,S	Small et al. (2007)
	<i>Hematodinium</i> sp.	NM; Georgia, USA	PCR,S,PA,DHPLC	Troedsson et al. (2008)
	<i>Hematodinium</i> sp.	Low infected; Coastal Maryland & Virginia, USA	HS,PCR	Nagle et al. (2009)
	<i>Hematodinium</i> sp.	June; Virginia, USA	HS,H,PCR,S	Pagenkopp Lohan et al. (2012)
	<i>Hematodinium</i> sp.	December; Carolina, USA	MA,PCR	Parmenter et al. (2013)
	<i>H. perezi</i> sp.	Spring; Atlantic and Gulf, USA	PCR,S	Hanif et al. (2013)
	<i>H. perezi</i> sp.	NM; Virginia, USA	HS	Butler et al. (2014)
	<i>H. perezi</i> sp.	NM; Louisiana, USA	PCR	Sullivan et al. (2016)
	<i>Hematodinium</i> sp.	NM; Adana, Turkey	HS	Aldik and Cengizler (2017)
	<i>H. perezi</i> sp.	Fall and summer; Maryland Coastal Bays, USA	I,PCR	Lycett et al. (2018)
	<i>Hematodinium</i> -Like	Low infected; Adriatic Sea, Italy	HS	Patrizia and Giorgio (2018)
	<i>H. perezi</i> sp.	Summer and fall; Coastal Bays of Virginia, USA	PCR	Small et al. (2019)
<i>Cancer borealis</i> sp.	<i>Hematodinium</i> sp.	All year; NY Bight, USA	H	MacLean and Ruddell (1978)
<i>Cancer irroratus</i> sp.	<i>Hematodinium</i> sp.	NM; NY Bight, USA	H	MacLean and Ruddell (1978)

Table 1 (continued)

Host	<i>Hematodinium</i> species	Season/month; reported places	Morphological methods used	Reference (s)
<i>Chionoecetes bairdi</i> sp.	<i>Hematodinium</i> sp.	Summer; East panhandle of Alaska, USA	HS,H,U	Meyers et al. (1987)
	<i>Hematodinium</i> sp.	Nearly equal; Southeast Alaska, USA	MM,HS	Meyers et al. (1990)
	<i>Hematodinium</i> sp.	Summer; Southeast Alaska, USA	HS,DS	Eaton et al. (1991)
	<i>Hematodinium</i> sp.	Summer; Auke Bay, Alaska, USA	HS	Love et al. (1993)
	<i>Hematodinium</i> sp.	NM; Alaska, USA	HS,PCR	Hudson and Adlard (1994)
	<i>Hematodinium</i> sp.	1988; Alaska, USA & Bering Sea, Russia	HS,H	Meyers et al. (1996)
	<i>Hematodinium</i> sp.	NM; Kodiak, USA	HP	Urban and Byersdorfer (2002)
	<i>Hematodinium</i> sp.	NM; Southeast Alaska, USA	PCR,S	Crandall et al. (2022)
<i>Cancer pagurus</i> sp.	<i>Hematodinium</i> sp.	Winter; English Channel, Iroise Sea, Bay of Biscay & West off Scotland	HP	Latrouite et al. (1988)
	<i>Hematodinium</i> sp.	NM; South Coast of Guernsey, UK Channel Islands	H,U,I,PCR	Stentiford et al. (2002)
	<i>Hematodinium</i> sp.	Autumn; Scottish West Coast, UK	PCR, S	Hamilton et al. (2009)
	<i>Hematodinium</i> sp.	Winter/Late autumn; Southwest, North & Southeast of Ireland	HS	Ni Chualáin et al. (2009)
	<i>Hematodinium</i> sp.	NM; Wales & Mumbles Head, UK	HP,H,PCR,PA	Smith et al. (2013)
	<i>Hematodinium</i> sp.	November; Coast of Malin Head, Ireland	HS,H,PCR	Ni Chualáin and Robinson (2011)
<i>Necora puber</i> sp.	<i>Hematodinium</i> sp.	Spring; South Wales, UK	HP,H	Smith et al. (2015)
	<i>Hematodinium</i> sp.	Winter; Brittany, France	HM,U	Wilhelm and Boulo (1988)
	<i>Hematodinium</i> sp.	Autumn; Brittany, France	HS,H,U	Wilhelm and Mialhe (1996)
	<i>Hematodinium</i> sp.	NM; English Channel	H,U	Stentiford et al. (2003)
	<i>Hematodinium</i> sp.	Autumn; West Coast of Scotland	PCR,S	Hamilton et al. (2009)
	<i>Hematodinium</i> sp.	NM; Galway Bay, Castle-Townbere & Howth, Ireland	H,PCR	Collins et al. (2022)
<i>Trapezia areolata</i> sp.	<i>Hematodinium</i> sp.	NM; Great Barrier Reef, Australia	HP,U	Hudson et al. (1993)
<i>Trapezia coerulea</i> sp.	<i>Hematodinium</i> sp.	All year, low infected; Great Barrier Reef, Australia	HP	Hudson et al. (1993)
<i>Portunus pelagicus</i> sp.	<i>Hematodinium australis</i> n. sp	All year, low infected; Moreton Bay, Australia	HS,H,U	Hudson and Shields (1994)
<i>Scylla serrata</i> sp.	<i>Hematodinium</i> sp.	All year, low infected; Queensland, Australia	HS	Hudson and Lester (1994)
	<i>Hematodinium</i> sp.	NM; Coastal Eastern China	HP,U,PCR,S	Xu et al. (2007a, 2007b)
	<i>Hematodinium</i> sp.	NM; Guangdong, China	HP,PCR	Li et al. (2008)

Table 1 (continued)

Host	<i>Hematodinium</i> species	Season/month; reported places	Morphological methods used	Reference (s)
<i>Callinectes similis</i> sp.	<i>Hematodinium</i> sp.	NM; Georgia, USA	HP	Messick and Shields (2000)
	<i>Hematodinium</i> sp.	Low infected; Georgia, USA	HP,H	Sheppard et al. (2003)
<i>Chionoecetes opilio</i> sp.	<i>Hematodinium</i> sp.	NM; Newfoundland, Canada	HP,U	Taylor and Khan (1995)
	<i>Hematodinium</i> sp.	1999; Newfoundland	MM	Dawe et al. (2001)
	<i>Hematodinium</i> sp.	NM; Conception Bay, Newfoundland	MM,HS	Pestal et al. (2003)
	<i>Hematodinium</i> sp.	NM; Conception & Bonavista Bays, & Avalon, Newfoundland	MM,HS	Shields et al. (2005)
	<i>Hematodinium</i> sp.	NM; Conception Bay, Newfoundland Canada	MM	Shields et al. (2007)
	<i>Hematodinium</i> sp.	NM; Northeast coast of Newfoundland	H	Wheeler et al. (2007)
	<i>Hematodinium</i> sp.	NM; Sisimuit, West Greenland	MM,PCR,S	Eigemann et al. (2010)
	<i>Hematodinium</i> sp.	NM; White Bay, Newfoundland	HP,U	Gaudet et al. (2015)
	<i>Hematodinium</i> sp.	Fall; Bering & Chukchi Seas, Russia	HS	Ryazanova et al. (2016)
	<i>Hematodinium</i> sp.	NM; Virginia, USA	HS,PCR,S	Pagenkopp Lohan et al. (2012)
<i>Chionoecetes tanneri</i> sp.	<i>Hematodinium</i> sp.	September; British Columbia, Canada	HP, H	Bower et al. (2003)
<i>Libinia emarginata</i> sp.	<i>Hematodinium</i> sp.	October; GA, USA	HP,H,PCR,S	Sheppard et al. (2003)
	<i>Hematodinium</i> sp.	NM; Virginia, USA	HS,H,PCR,S	Lohan et al. (2012)
<i>Neopanope sayi</i> sp.	<i>Hematodinium</i> sp.	September; GA, USA	HP, H	Sheppard et al. (2003)
<i>Menippe mercenaria</i> sp.	<i>Hematodinium</i> sp.	Low infected; GA, USA	HP,H,PCR,S	Sheppard et al. (2003)
	<i>Hematodinium</i> sp.	NM; West coast of Greenland	PCR	Eigemann et al. (2010)
<i>Pagurus bernhardus</i> sp.	<i>Hematodinium</i> sp.	NM; Clyde Sea, UK	PCR,S	Small et al. (2007)
	<i>Hematodinium</i> sp.	Spring; Clyde Sea, Scotland	PCR,S	Hamilton et al. (2009)
	<i>Hematodinium</i> sp.	NM; The Sound, Denmark	MM,PCR,S	Eigemann et al. (2010)
<i>Portunus armatus</i> sp.	<i>Hematodinium</i> sp.	NM; Shark Bay, Western Australia	HS	Musgrove and Slattery (2008)
<i>Paralithodes camtschaticus</i> sp.	<i>Hematodinium</i> sp.	Summer; Sea of Okhotsk, Russia	HP,H	Ryazanova (2008)
<i>Paralithodes platypus</i> sp.	<i>Hematodinium</i> sp.	Summer; Sea of Okhotsk, Russia	HP,H	Ryazanova (2008)
<i>P. trituberculatus</i> sp.	<i>Hematodinium</i> sp.	September; Coastal eastern China	HP,H,U	Xu et al. (2007b)
	<i>Hematodinium</i> sp.	NM; Shandong, China	HP,U,PCR,S,PA	Li et al. (2013)
	<i>Hematodinium</i> sp.	NM; Zhejiang, China	HS,H,U	Wang et al. (2015)
	<i>H. perezii</i> sp.	July; Shandong, China	HS,PCR	Huang et al. (2021)
<i>Pagurus prideaux</i> sp.	<i>Hematodinium</i> sp.	Low infected; Clyde Sea, Scotland	PCR,S	Hamilton et al. (2009)
<i>Munida rugosa</i> sp.	<i>Hematodinium</i> sp.	April; Clyde Sea, Scotland	PCR,S	Hamilton et al. (2009)
<i>Cancer magister</i> sp.	<i>Hematodinium</i> sp.	NM; Alaska, USA	HS	Meyers and Burton (2009)

Table 1 (continued)

Host	<i>Hematodinium</i> species	Season/month; reported places	Morphological methods used	Reference (s)
<i>Paralithodes camtschaticus</i> sp.	<i>Hematodinium</i> sp.	NM; Sea of Okhotsk, Russia	MM,H,HP,PCR,S	Ryazanova et al. (2010)
<i>Paralithodes platypus</i> sp.	<i>Hematodinium</i> sp.	NM; Sea of Okhotsk, Russia	MM,H,HP,PCR,S	Ryazanova et al. (2010)
<i>Lithodes couesi</i> sp.	<i>Hematodinium</i> sp.	NM; West Vancouver Island, Canada	PCR,S	Jensen et al. (2010)
<i>Chionoecetes angulatus</i> sp.	<i>Hematodinium</i> sp.	NM; West Vancouver Island, Canada	PCR,S	Jensen et al. (2010)
<i>Hyas coarcticus</i> sp.	<i>Hematodinium</i> sp.	NM; Eastern Bering Sea	PCR,S	Jensen et al. (2010)
<i>Libinia dubia</i> sp.	<i>Hematodinium</i> sp.	NM; Virginia, USA	HS,H,PCR,S	Pagenkopp Lohan et al. (2012)
<i>Pagurus pollicaris</i> sp.	<i>Hematodinium</i> sp.	NM; Virginia, USA	HS,PCR,S	Pagenkopp Lohan et al. (2012)
<i>Eurypanopeus depressus</i> sp.	<i>Hematodinium</i> sp.	NM; Virginia, USA	HS,PCR,S	Pagenkopp Lohan et al. (2012)
<i>Leptomithrax gaimardii</i> sp.	<i>Hematodinium</i> sp.	NM; Port Philip Bay, Australia	HP,PCR,,I	Gornik et al. (2013)
<i>Ovalipes australiensis</i> sp.	<i>Hematodinium</i> sp.	NM; Port Philip Bay, Australia	HP,PCR,,I	Gornik et al. (2013)
<i>Plagusia chabrus</i> sp.	<i>Hematodinium</i> sp.	NM; Port Philip Bay, Australia	HP,PCR,,I	Gornik et al. (2013)
<i>S. paramamosain</i> sp.	<i>Hematodinium</i> sp.	NM; Guangdong, China	HS,PCR,S	Wang et al. (2017)
<i>Hapalogaster grebnitzkii</i> sp.	<i>Hematodinium</i> sp.	Low infected; Sea of Okhotsk, Russia	MM,H,U,PCR	Ryazanova et al. (2018)
<i>Eriphia verrucosa</i> sp.	<i>Hematodinium</i> -Like	July; Acquatina Lagoon, Adriatic Sea, Italy	HS	Patrizia and Giorgio (2018)
<i>Helice tientsinensis</i> sp.	<i>Hematodinium</i> sp.	July; Huangdao, China	MM,HS,H	Huang et al. (2019)
		NM; Shandong, China	HS,PCR	Huang et al. (2021)
<i>Hexapanopeus angustifrons</i> sp.	<i>Hematodinium</i> sp.	All month, low infected; Maryland, North Carolina	HP	Messick and Shields (2000)
<i>Hemigrapsus penicillatus</i> sp.	<i>Hematodinium</i> sp.	Low infected; Shandong, China	HS,PCR	Li et al. (2021)
<i>Paralithodes brevipes</i> sp.	<i>Hematodinium</i> sp.	All month, low infected; Kamchatka, Russia	MM,H,U,PCR	Ryazanova et al. (2021)
<i>Helice wuana</i> sp.	<i>Hematodinium</i> sp.	All month, low infected; Shandong, China	HS,PCR	Li et al. (2021)
<i>Uca arcuate</i> sp.	<i>Hematodinium</i> sp.	September; Shandong, China	HS,PCR	Li et al. (2021)
<i>Macrophthalmus japonicus</i> sp.	<i>Hematodinium</i> sp.	Low infected; Shandong, China	HS,PCR	Li et al. (2021)
<i>Hemigrapsus takanoi</i> sp.	<i>H. perezii</i> sp.	Autumn; Shandong Peninsula, China	MM,H,PCR	Gong et al. (2023)

MM Morphological method including body color, pleopod diagnosis, carapace shape; HS hemolymph smear; HP hemolymph preparation; H histopathology; U ultrastructure; I immunoassay including ELISA, IFA, and Ecd-RIA; PCR polymerase chain reaction; S sequencing; PA phylogenetic analysis; PG population genetics; DHPLC denaturation high-performance liquid chromatography

efforts and mitigation methods for these effects can be further elaborated.

Apart from glycogen and glucose, it is known that the hormone of methylfarnesoate (MF) also plays a positive role in *Hematodinium*. Methyl farnesoate is a sesquiterpenoid hormone involved in physiological processes and

crustacean development (Riboni et al. 2022). The parasite does not have the essential isoprenoid pathway; therefore, they took isoprenoids from the host's hemolymph (Gornik et al. 2015). This MF hormone is suspected as a source of isoprenoids required by pathogens, and researchers suggested that the rise in MF hormone levels just before a

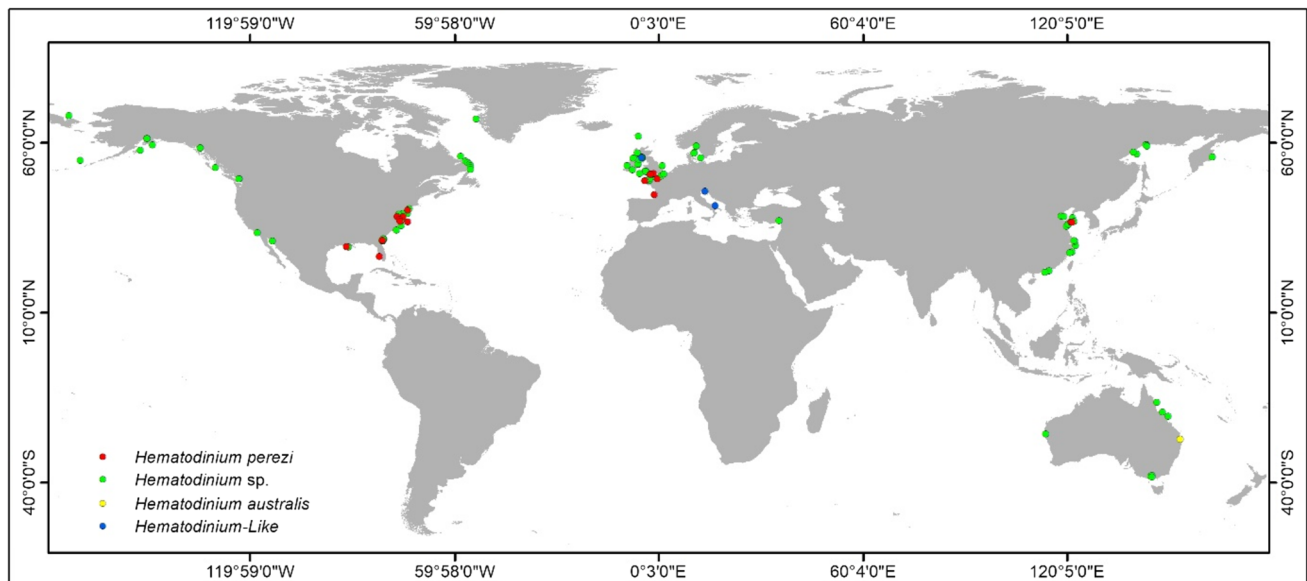


Fig. 1 The map exhibits the worldwide spread with variety of *Hematodinium* species. Note: *Hematodinium perezii* (red dot); *Hematodinium* sp. (green dot); *Hematodinium australis* (yellow dot); and *Hematodinium*-like disease (blue dot)

molt can promote growth and set off the development of dinospora.

Gross signs and pathogenicity

Fishing industries and host populations are both adversely affected by this *Hematodinium* invasion, which is comparable to viral illnesses in crustaceans (Shields et al. 2015). Infected crustaceans exhibit systemic signs such as lethargy and cloudy or milky hemolymph that does not clot in the periphery or around the internal viscera (Stentiford and Shields 2005). In addition, there are other microscopic symptoms, such as discolored carapace, although the results are inconsistent in each species. Carapace hyperpigmentation in infected hosts turned pink and cooked, as happened in *C. opilio*, *S. serrata*, *E. carinicauda*, and *C. bairdi* (Pestal et al. 2003); (Li et al. 2008); (Xu et al. 2010); (Ryazanova et al. 2021); however, the carapace color change was not found in other species, including *P. trituberculatus*, *C. maenas*, *P. brevipipes*, and in heavily infected *P. camtschaticus* and *P. platypus* (Ryazanova et al. 2010; Ryazanova et al. 2021); (Davies et al. 2019); (Li et al. 2023). (Ní Chualáin et al. 2009) also reported inconsistencies regarding the impact of this invasion, where they observed more than 4000 *C. pagurus* and found only less than 1% had a pink color change. Carapace hyper-pigmentation, in which the normally reddish carapace turns pink, has been observed in the arthrodia membranes of the following species: *N. norvegicus*, *Cancer bairdi*, *C. opilio*, and *C. pagurus* (Stentiford and Shields 2005). Crabs of the species *C. opilio* and *C. bairdi*

found in the USA have chalky meat that is soft and has an unpleasant flavor when cooked (Meyers et al. 1987), and at later stages of infection, the host may not be able to breathe, tissues may not be getting enough oxygen, and biochemical changes may deplete protein and glucose stores, leading to death (Stentiford and Shields 2005).

Diseases are classified according to how they alter the hemolymph, tissues, and organs. Illness caused by *Hematodinium* invasion is known by several names, including bitter crab syndrome (BCS), bitter crab disease (BCD), milky disease (MD), and yellow water disease (YWD). Pink crab disease (PCD), which is determined based on symptoms of carapace discoloration in the host, was also considered one of the *Hematodinium* infections (Stentiford et al. 2002); (Pestal et al. 2003); (Li et al. 2008); (Smith et al. 2015); (Ryazanova et al. 2021); however, these symptoms show inconsistent results in each species. PCD shows symptoms where hemolymph and muscle turn pink in color and have an uneven fleshy texture and a bitter taste. Additionally, this PCD also causes yellow discoloration of the arthroal membranes and genital pores, which was later revealed to be due to the consistency and cream color of the hemolymph (Stentiford et al. 2002), and hyperpigmentation, which is related to the host response to pathogens (Stentiford and Shields 2005). Moreover, (Rowley and Coates 2023) have described that it is not considered an actual shell disease if only changes in pigmentation occur due to host response; indeed, that the term shell disease refers to progressive carapace embrittlement due to the action of pathogenic enzymes.

BCD or BCS was first documented to occur in *C. opilio* and *C. bairdi* shrimp in the USA, resulting in mushy meat

and an unpleasant, bitter chalky taste when served (Eaton et al. 1991); (Meyers et al. 1987). Bitter-tasting flesh due to *Hematodinium* infection was also reported in velvet swimming crab *Necora puber* from French waters (Wilhelm and Mialhe 1996)). This phenomenon was also documented in the red king crab *Paralithodes camtschaticus*, and the blue king crab *Paralithodes platypus* in Russian waters (Ryazanova 2008). Generally, BCD is found in juvenile male crabs and immature female crabs due to their tendency to molt more frequently (Shields et al. 2005). Although BCD poses no health risk to humans, it does render contaminated crab meat unsellable due to its chalky texture and bitter taste (Crandall et al. 2022). The aspirin-like bitter taste of infected crabs is suspected to come from biochemical and physiological host changes due to prolonged, chronic infection with protozoa. Additionally, the droplets that emerge from the vegetative surface may also be an extracellular product of *Hematodinium* foamy cytoplasm (Meyers et al. 1987); (Stentiford and Shields 2005).

The history of PCD began when (Latrouite et al. 1988) reported the invasion of *Hematodinium*-like parasites in *C. pagurus* populations from the English Channel, the Irish Sea, the Bay of Biscay, and the west coast of Scotland. The symptoms include hemolymph and muscle, which are pink in color and have an uneven flesh texture and a bitter taste. (Stentiford et al. 2002) also found that *C. pagurus* crabs captured off the west coast of Cornwall and the island of Guernsey with carapace symptoms developed hyperpigmentation (pink) and yellow discoloration of the arthroal membranes and genital pores. This discoloration was later revealed to be attributable to the hemolymph's creamy consistency and color due to the parasite's intensity in large quantities. Cream-colored surfaces of organs and tissues are caused by parasite deposits (Stentiford et al. 2002).

MBS has also been reported as a result of a *Hematodinium* invasion. It was first documented to attack the blue swimming crab *P. trituberculatus* (Xu et al. 2007b), and the mud crab *S. serrata* (Xu et al. 2007a). The clinical signs displayed in this disease are generally similar to other diseases caused by *Hematodinium*, like BCD or PCD (Meyers et al. 1987); (Stentiford et al. 2002); (Taylor and Khan 1995), where their symptoms include sluggish host behavior, death during capture and transfer, darkening of the carapace, milky hemolymph, and chela muscle becoming liquid (Li et al. 2008). Meanwhile, (Kon et al. 2011) have also reported that MD with symptoms of darkening of the hemolymph and carapace also occurs in *C. opilio* snow crabs from the Sea of Japan, whereas yellow water disease (YWD) was introduced by *S. serrata* cultivating farmers in Shanmen, Zhejiang, China, in September 2005. The symptoms caused are almost the same as those of SBM, notably whitened muscular tissue, milky fluid when the carapace is removed, and a slender body (Xu et al. 2007 Xu et al. 2007). It is known

that MBS/MD, YWD, and MSD are diseases caused by the invasion of *Hematodinium* and identified based on observations of changes in the color and texture of the hemolymph.

Milky shrimp disease is known among farmers in China as an illness caused by a discolored carapace and hemolymph that looks like milk. Because its gross clinical indications are similar to those of *P. trituberculatus* infected with *Hematodinium*, it is considered that *Hematodinium* is likewise an epidemic agent of MSD in *E. carinicauda* (Xu et al. 2010). Signs produced by this disease include shrimp that look lethargic, move little, and stay in shallow water; a carapace with a "ripe" color characteristic of crustaceans when cooked; and hemolymph, which is like milk and does not clot. These symptoms are comparable to those of MBS disease reported in *S. serrata* in southern China (Li et al. 2008). However, there are additional symptoms, such as white muscles on walking legs and claws in infected shrimp (Xu et al. 2010).

Environmental drivers of disease

Several factors affect the epidemiology of *Hematodinium*, one of which is environmental factors such as salinity and temperature, where it is known that this parasite tends to infect hosts in high-salinity waters (Barbosa et al. 2023). Shields and Overstreet (2007) reported that infections were rarely found in salinities less than 18 psu, and even no infection of this parasite was found in salinities less than 11 psu. Low salinity, such as 15 psu and lower, will make dinospores inactive and thus unable to transmit to new hosts (Huchin-Mian et al. 2017), although this situation does not always guarantee to eliminate infection and the mortality rate of infected crabs (Coffey et al. 2012); (Huchin-Mian et al. 2018). For instance, (Li et al. 2008) reported the occurrence of *Hematodinium* in *S. serrata* cultured at low salinity (i.e., <9 psu). Meanwhile, (Sullivan et al. 2016) reported the detection of *H. perezi* sp. in blue crabs in Louisiana, both in high-salinity offshore waters and low-salinity marshes. Besides, (Dawe et al. 2010) also found that there were almost no *Hematodinium* infections in water with a salinity below 3 psu and in the saltiest waters, which are more than 35 psu. Generally, high salinity ranging from 30 to 34 psu will make blue crabs highly susceptible to parasite infection; however, (Gandy et al. 2015) argue that increased salinity alone is not sufficient to modulate increased parasite prevalence.

Other than salinity, increased temperature is also an important driver for transmission modulation of *H. perezi* sp. in hosts; as the temperature goes up, dinospores are released to infect new hosts (Gandy et al. 2015); (Shields et al. 2015); (Huchin-Mian et al. 2017). (Sheppard et al. 2003) showed regular sharp peaks in prevalence in late autumn, followed by a rapid decline in winter and a moderate increase in

spring, with a small peak in late spring or early summer. (Small et al. 2019) also reports a high prevalence of *Hematodinium* infection during the autumn and a decrease to zero starting in the winter. (Shields et al. 2015) explained that cold temperatures limit *H. perezi* sp. infection due to the crab's inability to adapt to cold temperatures.

Several studies have revealed the conditions of temperature and salinity that support the invasion of this parasite. (Messick and Shields 2000) found the highest prevalence of *Hematodinium* at salinities of 26–30 psu and when the water temperature was > 25 °C. (Huchin-Mian et al. 2018) reported that deaths due to *H. perezi* sp. infection occurred at a temperature of 25 °C and a salinity of 30 psu, which is also the optimal value for *Hematodinium* to develop and release dinospores to infect new hosts. (Shields et al. 2015) argued that the prevalence and intensity of *Hematodinium* decrease at lower temperatures. It is important to note that crustaceans are ectothermic animals, and temperature fluctuations greatly affect their physiological and immunological processes in responding to pathogens (Li et al. 2023). For *Hematodinium*, temperature determines its life cycle (Huchin-Mian et al. 2018). (Chen et al. 2023) reported that temperature increases of 25 °C during late summer and early autumn strongly influence the acceleration of *Hematodinium* parasite reproduction in the wild. (Sullivan and Neigel 2017) reported that a temperature higher than 34 °C results in high mortality in infected crabs compared to uninfected crabs, which might be due to the increase in stress factors. Continuous scientific research is essential to protect and maintain the health of *Hematodinium* and other marine biodiversity.

Potential economic damages on crustacean production

This dinoflagellate parasite, *Hematodinium*, is thought to affect fisheries and host populations in a way that is similar to how viral diseases of crustaceans affect fisheries and host populations (Small et al. 2012), as they cause significant mortality in the host. In Alaska, in the early 1980s, there was 100% mortality in *C. bairdi* crab farming. It is estimated that industry players, such as companies, lost up to \$250,000 due to *C. bairdi* crabs needing to be more marketable due to the bitter taste of crab meat infected with *Hematodinium* (Meyers et al. 1987). *C. sapidus* crab from the coastal bays in Maryland and Virginia had a prevalence rate of 70 to 100% in 1992, with implications for losses of as much as \$1.5 million during major epizootics (Messick 1994) or about \$250,000 to \$500,000 annually in Virginia (Messick 1994); (Pestal et al. 2003). Along the east coast and the Gulf coast of the USA, in 2011 to 2012, the *H. perezi* parasite-infected *C. bairdi* with a prevalence rate of up to 75 to 80%, which

caused the crabs to suffer illness and eventually die due to infection from the parasite (Shields et al. 2015).

In another reported case in Scotland, the lobster fisheries sector for *N. norvegicus* loses £2 to 4 million annually due to invasion by parasites such as *Hematodinium* (Field and Appleton 1995); (Field et al. 1992)). One of the sources of this loss is that around 30% of the total lobster caught each season is discarded because it shows a discolored carapace, which is often associated with a symptomatic or asymptomatic *Hematodinium* infection. This discoloration is considered a decrease in quality, which does not suit their target market, which prefers quality over volume (Albalat et al. 2016). In France, the capture fishery for *N. puber* experienced a production decline of up to 96% from 1984 to 1988 due to *Hematodinium* infection, with a prevalence of 80% (Wilhelm and Mialhe 1996). In the Bristol Channel (UK), juvenile *C. pagurus* was reported to cause a significant reduction in stocks with an infection rate of up to 30% (Smith et al. 2015). Frequent outbreaks of *Hematodinium* infections have caused economic losses for aquaculture in China. In 2005, mud crabs in southern China suffered from an epizootic caused by the invasion of *Hematodinium*, resulting in mortality rates above 60%, which caused huge losses from an economic perspective (Li et al. 2008).

The cultivation of white ridgetail shrimp *E. carinicauda* has been hampered since August 2008 due to MSD disease, which resulted in up to 100% mortality in Zhoushan, Zhejiang Province, China (Xu et al. 2010). Recent studies reported that *H. takanoi* sp. is a new host for *Hematodinium* with a fairly high prevalence, reaching 20% in Weifang, China (Gong et al. 2023). Currently, crustaceans' cultivation is dominated by shrimp, whose activities are often carried out in the coastal areas of tropical and subtropical countries (Nguyen et al. 2020). China is no exception, whereas other Asian countries and Latin America have made shrimp farming an important foreign exchange earner for the country (Tacon 2020), so it can be concluded that the invasion of *Hematodinium* has a very big impact and has the potential to disrupt shrimp farming activities, which will ultimately harm producers and consumers in terms of the economy and supply of nutritional sources.

Mode of disease transmission and control methods

Although the exact way by which the parasite invades its host remains unknown, it is suspected that crustaceans are more susceptible to infection after the molting period (Shields et al. 2007; Shields et al. 2005). When the cuticle is thin and porous, invaders use the inguinal pathway to access hosts. The parasite penetrates the underlying tissues and the main body space, called the hemocoel, and multiplies in the

heart, hepatopancreas, connective tissue, and hemolymph (Davies and Rowley 2015); (Rowley et al. 2015); (Stentiford and Shields 2005).

Another entry point for dinospores is through the gills, or integumentary gills (Hanif et al. 2013). Dinospores are the product of sporulation in *Hematodinium* (Huchin-Mian et al. 2018). Both micro- and macro-dinospores are released into the environment through the host's gills, feces, and urine. Nevertheless, it is most likely released in large quantities during tissue decomposition after the host dies (Meyers et al. 1987). The release of dinospores into the environment causes the water to change color. In the absence of a host, dinospores *Hematodinium* can live for up to seven days in the water column (Huchin-Mian et al. 2018); (Li et al. 2010). In contrast, (Meyers et al. 1987) reported that dinospores was found to live in sterile saltwater for 73 days after being released from tanner crabs. Based on the above statement, we believe that *Hematodinium* dinospores has a brief aquatic phase in its life cycle. Until now, the transmission routes of *Hematodinium* were not fully understood, which hindered understanding and precautions against this epizootic. However, it is known that waterborne transmission of *Hematodinium* has occurred, and it may originate from dinospores free-living in water and suspended sediments.

Several studies have established this transmission by confirming the presence of *Hematodinium* in the water column using Internal Transcribed Spacer region 1 (ITS1) targeting primers of the *Hematodinium* ribosomal RNA (rRNA) gene complex (Li et al. 2010); the PCR method found that both water and sediment were positive for *Hematodinium* (Frischer et al. 2006); (Pitula et al. 2012); ITS2-targeted testing used *Escherichia coli* carrying the *H. perezii* rDNA gene (Hanif et al. 2013); and detection of *H. perezii* DNA in the bay water column (Lycett and Pitula 2017). According to (Frischer et al. 2006), *C. sapidus* can be infected if it is kept in close proximity to infected crabs or if it is kept in an aquarium that previously housed crabs infected with dinospores. The recognition and spread of *Hematodinium* parasites through the water column is supported by (Huang et al. 2021), who found that *H. perezii* can be transmitted from *H. tientsinensis* to *P. trituberculatus*.

The latest transmission line has been reported by (Chen et al. 2023), which describes that an increase in temperature (e.g., late summer and early autumn) can support the range and natural transmission of *H. perezii*. A map of the natural transmission of *H. perezii* on a laboratory scale results in quick transmission through water media at 25 °C (Chen et al. 2023). Meanwhile, Shield et al. (2007) speculated that an increase in temperature causes more molting activity to occur at once, thereby making them more vulnerable to infection. Others have also noted a positive correlation between temperature and the presence of parasites, particularly when it is associated with the current global warming

trend towards the transmission of infection, the distribution of parasites in places, and the ranges of other species and fishing grounds.

Crabs, as they grow older and larger, become increasingly hostile and cannibalistic. For this reason, there are situations in which he will consume individuals of his own species that are significantly younger than him. When put in the context of its involvement in the spread of *Hematodinium*, this becomes an intriguing finding. (Walker et al. 2009) reported that 73% of *C. sapidus* crabs were infected with *Hematodinium* after 16 h of feeding on contaminated tissue fragments; however, results contradicted those of (Li et al. 2011), where only 2 out of 120 crabs were infected. These findings may be due to low exposure to infection, where the infected crabs only carried a low intensity of infection before the experiment. Hudson and Shield (1994) revealed the challenges of cannibalistic transmission, reporting that *P. pelagicus* remained uninfected despite being given *Hematodinium*-infected tissue fragments. Nonetheless, *Hematodinium* was successfully transmitted to *C. sapidus* crabs through feeding on diseased tissue, as discovered by (Sheppard et al. 2003). In a separate study, *N. norvegicus* was reported to avoid eating infected tissue samples, highlighting the need for more investigation into the possible causes of this behavior (Hamilton 2007). Research by (Huang et al. 2021) argued that no pathogen transmission in the form of *Hematodinium*-infected tissue was observed when crabs were fed.

It has also been claimed that *Hematodinium* can be transmitted between hosts through sexual contact. During copulation, female lobsters undergo a skin change before males do, leaving their cuticles vulnerable, as argued by (Field et al. 1992). To mate with males who have hard shells, these requirements are required. There is a chance that *Hematodinium* could use this to get access to the host body. This may occur if larger crabs have more offspring, increasing the chances of dinospore migration. In the seminal fluid near the spermatozoa orifice, (Meyers et al. 1996) discovered many vacuolate, vegetative parasites. While *Hematodinium* in the injured crab's body might cause passive infiltration of the testes, sexually transmitted parasites can be passed on actively from one crab to another. Passive infiltration into the trauma area, including seminal fluid and potentially afflicted tissue, can result in the host's death, rendering him incapable of having sexual relations. However, it is necessary to re-evaluate this mode of transmission and provide additional data to confirm this hypothesis.

Conclusion and future perspectives

Although there are still many mysteries about *Hematodinium*, especially in bio-ecology, the authors convey several future perspectives in the field of research and

management that have the potential to be studied. One potential strategy is to employ RNA interference (RNAi) to pinpoint genes in the parasite's genome that can be silenced to disrupt the parasite's life cycle and impede the spread of infections. By comprehending these molecules, scientists can enhance their understanding of how *Hematodinium* interacts with its host in order to choose potential therapeutic targets. Moreover, understanding the epidemiology of *Hematodinium* which includes the distribution and prevalence of the parasite in different environments and species, as well as the influence of environmental factors on the transmission and survival of *Hematodinium*, are also crucial factors to be looked into. In addition, developing more efficient diagnostic instruments that are rapid, precise, and highly responsive, such as the visual loop-mediated isothermal amplification (LAMP) protocol, nanoparticle-based diagnostic approaches, the recombinase polymerase amplification (RPA), and micro-RNA (miRNA) are also important, so that farmers can promptly and precisely identify *Hematodinium* infections and implement necessary measures to prevent and control the disease from spreading. Additionally, the cultivation of crustacean species with higher resistance to this parasite by improving the non-immunological responses to the host immunity can also serve as a preventative measure. In a nutshell, collaboration among scientists, industry stakeholders, fishermen, and policymakers is essential to attaining these objectives.

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